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NON-IDENTICAL COGNATES YIELD FACILITATION IN TRANSLATION – DOES THE WAY FOREIGN VOCABULARY IS LEARNED AFFECT ITS PROCESSING?

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ABSTRACT

Cognates (words sharing form and meaning across languages, e.g. Polish–English FILM–FILM (identical cognates) or TUNNEL–TUNEL (non-identical cognates)) are processed faster than single-language words (*cognate facilitation effect*), which is modulated by a number of factors. Here, we extended the study reported by Lijewska and Chmiel (2015) and tested the influence of learning experience on non-identical cognate processing in a translation task in 2 experiments with Polish-German–English trilinguals and with German–English bilinguals. The trilinguals learned English and German via Polish, whereas the bilinguals learnt English via German. They translated Polish–English non-identical cognates (MUSZTARDA–SENF–MUSTARD), German–English non-identical cognates (TRAWA–GRAS–GRASS) and controls from English into Polish and into German (trilinguals) or only into German (bilinguals). We found no evidence for the influence of learning experience on cognate processing but the novel finding was a reliable facilitation effect obtained when only non-identical cognates were tested in a translation task with trilinguals (in the accuracy data) as well as with bilinguals (in translation latencies). Additionally, the reported study pointed to the important role of language proficiency in processing.

KEYWORDS: Non-identical cognate; trilingual; bilingual; learning experience; facilitation.

1. Introduction

In the vast majority of bilingual studies cognates (words sharing form and meaning across languages, e.g. FILM–FILM in Polish and English (Dijkstra et al. 2010)) have yielded a processing advantage (*cognate facilitation effect*). They were recognised (Brenders et al. 2011; Dijkstra et al. 1999; Lemhöfer et al. 2004, 2018; Peeters et al. 2013; Temnikova and Nagel 2015), translated

(Francis et al. 2014; Francis and Gallard 2005; de Groot et al. 1994; Van Hell 2005), read (Duyck et al. 2007; Libben and Titone 2009; Pivneva et al. 2014; Van Assche et al. 2009, 2011) and produced (Blumenfeld et al. 2016; Costa et al. 2000; Hoshino and Kroll 2008) faster and with greater accuracy than single-language words (non-cognate controls, e.g. the Polish–English translation pair STÓŁ-TABLE). This processing advantage has been typically interpreted as evidence for the co-activation of all languages known to the speakers during lexical access, i.e. when words are being located and activated in memory (Dijkstra et al. 2010, 2018; Dijkstra and Van Heuven 2002; Van Hell and de Groot 1998). The cognate advantage has been found to be modulated by a number of factors including the size of the cross-language word form overlap in cognates, (Comesaña et al. 2012, 2018; Dijkstra et al. 2010), their concreteness (Ferré et al. 2017; Van Hell and de Groot 1998) or grammatical category (Bultena et al. 2014; Van Assche et al. 2013), as well as the semantic bias of the sentence in which they appear (Libben and Titone 2009; Van Assche et al. 2011; Van Hell and de Groot 2008), participants' L2 proficiency (Bultena et al. 2014; Kaur 2017; Kootstra et al. 2012; Nakayama et al. 2013) or task demands (Dijkstra et al. 2010; Lemhöfer and Dijkstra 2004) (for reviews see Lauro and Schwartz 2017; Otwinowska 2015). A number of factors modulating cognate advantage have been identified in the course of bilingual research. In turn, the few studies testing cognate facilitation conducted with trilingual participants (Lemhöfer et al. 2004; Lijewska and Chmiel 2015; Poarch and Van Hell 2012, 2014; Szubko-Sitarek 2011, 2014; Van Hell and Dijkstra 2002) have helped to extend those findings so as to provide a better understanding of the mechanisms underlying cognate facilitation in particular and non-native language processing in general. The present study aims to contribute to this research by investigating the role of yet another factor in cognate processing by trilingual speakers. Namely, we aim to test whether the cognate facilitation effect is more robust when two languages sharing a cognate have been concurrently used during language learning relative to the situation when they were not used concurrently in learning (in the present study this factor will be referred to as *learning experience*). Furthermore, earlier bilingual studies have shown that non-identical cognates are sensitive to stimulus list composition in a lexical decision task (LDT) (Comesaña et al. 2015). When identical cognates were removed from the stimulus list, non-identical cognates were processed slower rather than faster than matched non-cognates (cognate inhibition). We aim to extend those findings to a translation task, where non-identical cognates (without any identical cognates in the stimulus list) will be translated by bilingual and trilingual speakers.

Many bilingual studies have shown that cognate facilitation is dependent on the type of cognates used as stimuli. Identical cognates (e.g. FILM-FILM in German and English), especially when they are concrete nouns, tend to yield a robust cognate advantage (Bultena et al. 2013, 2014; Dijkstra and Van Heuven 2002; Ferré et al. 2017; Van Hell and de Groot 2008). However, the size of this advantage decreases when the cross-language form overlap (phonological and orthographic) is incomplete, as is the case for non-identical cognates; e.g. BLUT-BLOOD in German and English (Comesaña et al. 2012, 2015; Dijkstra et al. 2010; Van Assche et al. 2011). Importantly, studies with trilingual speakers (Lemhöfer et al. 2004; Lijewska and Chmiel 2015; Szubko-Sitarek 2011, 2014) have extended these findings. The studies revealed than not only the size of the cross-language overlap affects the magnitude of the obtained advantage, but also the status of the languages (native vs. foreign) sharing a given cognate plays a role in the observed effects.

Szubko-Sitarek (2014) asked a group of Polish–English–German trilinguals to perform a LDT in their third language (L3) – German, wherein she compared processing patterns for two types of double cognates (i.e. cognates shared by two languages). Namely, she used Polish–German cognates, English–German cognates and German-only controls. This was the first study to test cognates shared by two foreign languages (English and German) next to the typically tested cognates shared by the native and a foreign language (Polish – L1 and German – L3). Szubko-Sitarek found evidence for a reliable cognate advantage only for Polish-German (shared across L1 and L3) cognates as the English–German ones (shared across L2 and L3) were not processed significantly differently from German-only (L3) control words.

Lijewska and Chmiel (2015) found similar evidence in a translation task where Polish–German–English trilinguals were asked to translate German–English double cognates (shared by L2 and L3) and English (L3) controls embedded at the end of two types of sentences. They were either semantically-constraining (high-context) sentences, (as in: On sunny days they picnic sitting on the green GRASS), or semantically-non-constraining (low-context) ones (as in: In the background she painted the GRASS). The participants were asked to silently read the sentences and to overtly translate the critical words (cognates and controls) from English (L3) into Polish (L1) and into German (L2). Apart from the significant context-constraint effects, the analysis revealed no significant cognate facilitation in translation from English (L3) into German (L2) but a significant cognate advantage in translation from English (L3) into Polish (L1). The authors hypothesized that this counterintuitive pattern of cognate facilitation (significant cognate advantage for German–English cognates

obtained while translating from English into Polish but not when translating from English into German) might stem from participants' learning experience as they all had been learning English (L3) via Polish (L1), not German (L2). As a consequence of such a learning experience, strong associative links formed between English words and their Polish equivalents but not between the English words and their German equivalents. This, in turn, lead to the observed, somewhat surprising, pattern of cognate facilitation. The hypothesized account of the recorded results was put forward by Lijewska and Chmiel in line with the combined assumptions derived from the Revised Hierarchical Model (RHM) (Dijkstra et al. 2010; Kroll et al. 2010; Kroll and Stewart 1994) and the Multilingual Interactive Activation (MIA) model (Dijkstra 2003). In line with MIA, Lijewska and Chmiel proposed that, when German–English cognates were translated into Polish, their overlapping lexical representations in English and in German were co-activated during recognition. Next, in line with RHM, the co-activated representations sent joint activation to their Polish translation equivalents via lexical links. In turn, these joint jolts of activation gave the Polish equivalents a head start in translation. Since non-cognates could not benefit from such a joint activation, they were translated slower than the cognates, which surfaced as the facilitation effect for German–English cognates during English–Polish translation. In contrast, the counterintuitive lack of cognate facilitation for German–English cognates during English–German translation the authors ascribed to weak lexical links between German and English lexical representations. The weakness of the links, they claimed, was a result of prior learning experience, as German and English had not been concurrently used by their participants during foreign language learning. In consequence, as Lijewska and Chmiel postulated, the representations of German–English cognates could not be activated via the lexical links (due to their weakness), thus translation from English into German operated via concepts and was in effect equally fast for cognates and non-cognates.

This line of reasoning is problematic for three reasons. Firstly, Lijewska and Chmiel combined assumptions derived from two separate models (RHM and MIA), only one of which (RHM) aims to model translation (the other is dedicated to visual word recognition). Secondly, Lijewska and Chmiel seem to posit that translation from English into Polish was lexically mediated, but translation from English into German was conceptually mediated. Such a scenario is highly unlikely as the translation was performed by advanced speakers of English and German. Therefore, in line with multiple evidence from previous research, the translation was most likely conceptually mediated in all directions (Christoffels et al. 2006; Francis and Gallard 2005; de Groot 1992; de

Groot et al. 1994; Kroll and Stewart 1994). Finally, it remains unclear why the supposedly weak lexical links between German–English cognates were strong enough to co-activate the overlapping representations in German and in English during English–Polish translation but not during English–German translation.

Therefore, in the present study, we aimed to replicate and extend the results reported by Lijewska and Chmiel (2015). Importantly, in their study, Lijewska and Chmiel did not include cognates shared by the native and the foreign language, which would enable them to more directly compare the processing patterns of both types of cognates (those that are shared by the concurrently learned languages and those that are not) and to verify the claim that cognates shared by the two languages concurrently used during learning (hence strongly linked in the mental lexicon), but not others, yield significant cognate facilitation across conditions. The present study aims to fill this gap. Moreover, the present research is based solely on the assumptions derived from RHM, which have been extended to trilinguals (see Fig 1).

Furthermore, the stimulus list used by Lijewska and Chmiel (2015) included identical and non-identical cognates, which might have influenced the obtained results. Comesaña et al. (2015) showed that, in an LDT, such stimulus list composition led to significant cognate facilitation. However, once identical cognates were removed from the list, the observed pattern for non-identical cognates changed into inhibition (i.e. cognates were recognised slower than controls). Comesaña et al. claimed that when identical cognates were absent, the non-target language became more highly suppressed leading to the slower, rather than faster, processing of the non-identical cognates relative to controls. They posited that this cognate disadvantage was caused by the lateral inhibition of non-identical cognates competing for selection, which was not the case for single-language controls. To the best of our knowledge, a stimulus list containing only non-identical cognates and controls has never been tested in a translation task. Therefore, the aim of the two experiments reported below is also to attempt to investigate this issue for the first time.

2. The present study

The aim of the present study was to test the influence of learning experience on processing patterns of 2 types of non-identical double cognates – Polish–English ones and German–English ones as well as matched English control words. We tested those stimuli in a translation task with two participant groups. One group comprised Polish (L1) trilingual speakers of German (L2) and Eng-

lish (L3) learned via Polish (Experiment 1), the other group comprised L1 German bilinguals with L2 English learnt via German (Experiment 2). We selected a group of German–English bilinguals who knew no Polish but who learned English via German, in Austria, a German-speaking country, since it was impossible to find a control group of Polish speakers who would use German when learning English (Polish is typically used during foreign language instruction in Polish schools).

In the present study, we followed the assumptions derived from RHM that lexical representations of words in the mental lexicon are linked via lexical links which are associative in nature (Dijkstra et al. 2010; Kroll et al. 2010; Kroll and Stewart 1994). Therefore, items from languages concurrently used during learning were considered to be strongly connected via associative, lexical links, whereas others only very weakly. Consequently, taking into consideration the language profile of our two groups of participants, we predicted the following connections between representations in the mental lexicons of our participants. Since the trilingual participants learned their foreign languages via Polish (L1), we assumed that there would be strong lexical links between each of the foreign languages (German and English) and L1 but significantly weaker ones between the two foreign languages (for a graphical illustration, see Fig. 1).

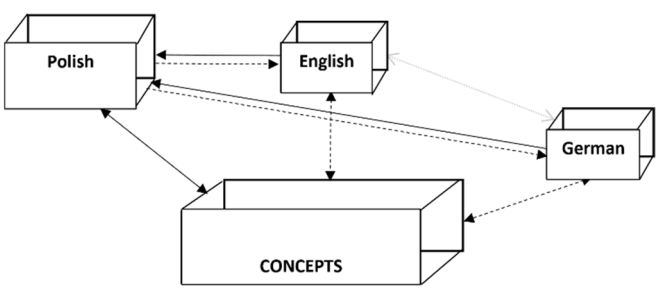

Fig. 1. Graphical illustration of the links between representations in the mental lexicon (in line with the assumptions of RHM (Kroll et al. 2010; Kroll and Stewart 1994) of a Polish trilingual speaker who used Polish during learning English and German as foreign languages. Full lines indicate strong links, dashed lines indicate weaker links, the pale dotted line indicates a very weak link. The pale dotted line has arrows on both ends to signify that, due to learning experience, the lexical link between two foreign languages is extremely weak and does not favour any direction. Boxes with language names indicate lexical representations in those languages, the box labelled "Concepts" signifies conceptual representations common to all languages.

In needs to be noted that, in line with past evidence from bilingual studies (Francis et al. 2014; de Groot 1992; Groot and Poot 1997) and from experiments with trilinguals (Francis and Gallard 2005), we assumed that translation in itself would be conceptually mediated in our participants (who were not beginner learners). In turn, following Dijkstra et al.'s (2010) explanation of the mechanics of the cognate facilitation effect within RHM, we assumed that this effect would operate along the lexical links between word forms. Consequently, we expected to find a robust cognate facilitation effect for Polish–English cognates (e.g. AKTOR–SCHAUSPIELER–ACTOR) relative to controls (e.g. POCIĄG–ZUG–TRAIN) in translation from English into Polish. This was based on the assumption that upon the presentation of the cue word ACTOR, its lexical representation would (along lexical links) co-activate the overlapping representation for AKTOR in Polish. In effect, this would lead to faster activation of the appropriate concept and speed up the retrieval of the Polish translation equivalent. Since only the cognate would benefit from the co-activation of two similar representations (ACTOR and AKTOR) via the strong lexical links (formed during language learning), translation of the non-cognate would be slower and the cognate facilitation effect would be observed.

This facilitation was expected to be slightly weaker in translation of the English word ACTOR into German (SCHAUSPIELER) than in the translation into Polish (AKTOR). Upon the presentation of the English cue word ACTOR for English–German translation, the Polish word form AKTOR would be co-activated via the strong lexical links and it would send an additional boost of activation to the semantic representation of the concept (via the semantic links). This, in turn, would facilitate the retrieval of the German translation equivalent SCHAUSPIELER. Such cognate facilitation would be absent when non-cognates were translated.

Importantly, the advantage for Polish–English cognates relative to non-cognates was expected to be weaker in English–German translation than in English–Polish translation. The underlying rationale was that during translation from English into Polish the co-activation of overlapping cognate forms across English and Polish would coincide with the translation direction, which would not be the case during the English–German translation, making the observed facilitation weaker.

In the case of English–Polish translation of the German–English cognates (e.g. PIWO-BEER-BIER) and non-cognates (e.g. COAT–PŁASZCZ–MANTEL) by the trilingual group, we expected the following sequence of processing steps. When the English cue word BEER would be presented, we expected only very small jolts of activation to travel to the German form BIER

since the lexical links between English and German word forms were expected to be very weak (as German and English had not been concurrently used during learning by our participants).

These small jolts of activation would next slightly raise the activation of the concept 'beer'. Consequently, thanks to the strong conceptual links between the concepts and L1 forms, the retrieval of the Polish (L1) equivalent PIWO would be facilitated. Such facilitation would not operate for non-cognates (no co-activation of overlapping forms possible) leading to the processing advantage for cognates relative to non-cognates during translation from English into Polish.

Finally, we predicted that when a German–English cognate (e.g. BEER) would be presented for translation into German, it would co-activate the German word form BIER via lexical links. Importantly, since the links were assumed to be weak (due to the lack of concurrent use of English and German during language learning), we predicted only a small amount of co-activation of the German lexical representation. The weakly co-activated forms (English and German) would, via the conceptual links, slightly facilitate the retrieval of the appropriate concept thereby slightly speeding up the retrieval of the German equivalent – BIER. Since the retrieval of the translation equivalent for the non-cognate would not be (even slightly) facilitated by the co-activation of the overlapping form, the process of translation would be slower for non-cognates than for cognates (cognate facilitation effect).

Importantly, we expected a slightly stronger cognate advantage for German–English (BEER–BIER) cognates relative to non-cognates (COAT) in the English–German translation than in the English–Polish translation. This prediction stems from the fact that during German–English translation only the source language (English) and the target language (German) representations would be co-activated and would next contribute to the expected facilitative effects. In contrast, during translation of German–English cognates and controls from English into Polish, the predicted cognate facilitation would stem from the co-activation of two lexical representations (one in English and one in German) but they would subsequently need to activate the appropriate concept thus leading to the retrieval of the Polish equivalent. These additional processing steps would most likely decrease the size of the observed facilitation.

As for the expectations concerning the results from Experiment 2, we expected robust facilitation for German–English cognates since those participants learned L2 English in the context of L1 German (hence strong lexical links exist between word forms in these two languages). As the participants

had no knowledge of Polish, we expected Polish–English cognates to be processed not significantly differently from the matched English-only items. As far as the secondary aim of the present experimentation is concerned, we predicted facilitation rather than inhibition for the non-identical cognates in our study. The prediction was not in line with Comesaña et al.'s (2015) findings, even though, similarly to Comesaña et al., we excluded identical cognates from our stimulus lists. The rationale for our prediction was based on the fact that task demands of a translation task are very different from those of an LDT (reported on in Comesaña et al.'s study). In contrast to LDT, where the non-target language can be highly suppressed to the benefit of performance, translation combines recognition in one language with production in the other so both languages need to remain active, although the relative levels of the co-activation may vary in time. Therefore, if the account of non-identical cognate disadvantage proposed by Comesaña et al. (2015) is correct, non-identical cognates tested in the present study should yield facilitation, not inhibition, due to different task demands.

2.1. Experiment 1

2.1.1. Participants

Forty-four Polish learners of German and English participated in Experiment 1. They were all students of Applied Linguistics at Adam Mickiewicz University in Poznan, Poland. Their curriculum included classes predominantly in German, fewer in English and only a few in Polish. The information concerning the participants' language background was collected via a set of questions adapted from LHQ 2.0 (Li et al. 2014). LexTALE (Lemhöfer and Broersma 2012) scores for English and German served as an independent measure of participants' vocabulary knowledge (a proxy for an estimate of proficiency). German and English had been both learned by the participants as foreign languages within the Polish education system, where Polish is typically used during foreign language instruction. Some of the participants started learning English before German, whereas others started learning German before English. Also, self-reported proficiency levels for English and German did not consistently indicate the most dominant foreign language. Consequently, we did not label the foreign languages as L2 or L3. In order to statistically control for possible effects of differences in the proficiency levels we added participants' LexTALE scores for German and English as continuous covariates in

the reported models. The summary of the participants' language profile is provided in Table 1.

Table 1. Basic characteristics of the participants in Experiments 1.

	German	English
Self-reported proficiency – Listening	5.5 (1.0)	5.8 (0.8)
Self-reported proficiency – Reading	5.5 (0.9)	5.1 (1.0)
Self-reported proficiency – Speaking	4.6 (1.2)	4.8 (1.1)
Self-reported proficiency – Writing	4.6 (1.0)	4.3 (1.3)
LexTALE score	61.8 (6.4)	62.1 (9.8)
Age of onset of foreign language learning	10.8 (3.7)	9.7 (4.0)
Age	22.4	

2.1.2. Materials

English words were used as stimuli - 36 double non-identical cognates and 36 matched noncognates as well as 36 non-cognate fillers. The aim of the fillers was to limit the noticeability of cross-language similarity in the task as we wanted to lower the likelihood of the use of any strategies by the participants. Out of 36 cognates 13 were similar across Polish and English (e.g. MUSZ TARDA–SENF–MUSTARD) and 23 were similar across German and English (e.g. TRAWA–GRAS–GRASS). The non-cognates (and fillers) were selected so as to be dissimilar to their translations in German and Polish. All of the items used in the experiment were concrete nouns to maximise the likelihood of obtaining significant cognate effects (Bultena et al. 2013, 2014). However, these strict requirements for stimulus selection lead to the imbalance in the number of stimuli across conditions. A list of stimuli is given in the Appendix.

Stimulus selection. The selection of stimuli was a two-step process. First, on the basis of a database from (Otwinowska 2015) as well as on the basis of German–English and PolishEnglish dictionaries, we selected nouns bearing similarity across English and Polish or across English and German. These items were next analysed orthographically with Van Orden's measure of similarity[1] (Van Orden 1987). Next, items identified as "likely cognates" were an-

[1] For a detailed explanation of Van Orden's algorithm for determining orthographic similarity see Van Orden (1987: 196)

alysed in terms of word length, frequency, dominant part of speech and on the basis of SUBTLEX-UK (Van Heuven et al. 2014). SUBTLEX-UK was selected over SUBTLEX-US since, in Poland, British rather than American English EFL materials dominate in education. Since our participants learned English within the Polish education system, we expected that SUBTLEX-UK would match their language experience better that SUBTLEX-US. Neighbourhood density counts were taken from CLEARPOND (Marian et al. 2012). For each of the possible cognates (nouns only), three matching non-cognate nouns were selected. Next, all these words were translated into German and Polish by a professional Polish translator of German and English. Translation pairs were next distributed across lists and distributed among 68 Polish learners of German and English for cross-language similarity and concreteness norming. Following Tokowicz et al. (2002), within the form similarity norming, the participants were asked to perform a combined rating and to decide (on a 7-point Likert scale 1 = dissimilar; 7 = very similar) how similar a given pair of words is, simultaneously with respect orthography and phonology. Within the semantic equivalence norming, the participants were asked to decide how similar with respect to meaning the pairs of words are (on a 7-point Likert scale 1 = dissimilar; 7 = identical). In order to avoid any carry-over effects, the lists for similarity norming were created in such a way that in each list the set of word pairs for form similarity norming was different from the set of words pairs for semantic equivalence norming. In a separate norming, we asked a different group of participants to assess the concreteness of the English words (on a 7-point Likert scale 1=abstract; 7=concrete). The concreteness norming was conducted online via the Instantly surveying programme (Instantly Inc. 2008). None of those who rated items during the norming sessions participated later in Experiment 1. On the basis of the results from the norming studies, we selected concrete nouns used as stimuli in Experiments 1 and 2. We considered similarity ratings a better measure of cross-language similarity than Van Orden's orthographic similarity scores since, during rating, the combined (phonological and orthographic) cross-language overlap was assessed.

Care was taken to minimise the orthographic similarity of non-cognates also across Polish and German even though no Polish-German translation was included in the experimentation. It was not always possible to completely eliminate non-cognate items with only a small degree of overlap across Polish and German. This was due to the fact that while selecting non-cognates they had to be matched to cognates with respect to a number of word characteristics, including potential familiarity to the participants. Since during the experimental task only English words were displayed, and the participants were

asked to translate them either into Polish or into German, but never from Polish into German nor vice versa, we decided to use the selected items despite the small degree of overlap in the non-target direction in a few of the non-cognate items. A t-test analysis of non-cognate items with respect to the similarity scores revealed that non-cognates were as dissimilar across Polish and German as they were across Polish and English ($p>.05$).

The MANOVA test, which was used to analyse the stimuli in terms of possible differences in their word level characteristics, revealed that Polish–English cognates as well as German–English cognates did not differ from their non-cognate controls with respect to the mean log frequency (Zipf-values), word length, orthographic and phonological neighbourhood density in English and in German, nor with respect to the concreteness ratings ($p>.05$). However, Polish–English cognates and non-cognates were significantly longer and less frequent than German–English cognates and non-cognates ($p<.05$). Of note, in order to statistically control for the influence of these differences, we added word length and frequency scores as covariates in the LME models. Most importantly, the MANOVA revealed that all stimulus types were significantly different from each other (all $ps<.01$) with respect to the rated cross-linguistic form similarity (in orthography and phonology combined). A summary of stimulus characteristics is presented in Table 2.

Table 2. A summary of stimulus characteristics (mean values; SD in parentheses). POL = Polish, ENG = English, GER = German, OND = orthographic neighbourhood density, PND = phonological neighbourhood density.

Stimulus type	POL–ENG				GER–ENG			
	cognates		non-cognates		cognates		non-cognates	
Length in letters	7.00	(2.1)	6.92	(2.0)	4.74	(1.3)	4.74	(1.3)
Frequency in Zipf-values	4.08	(0.5)	4.07	(0.5)	4.74	(0.6)	4.71	(0.5)
Rated form similarity	5.15	(0.8)	1.09	(0.2)	6.04	(0.3)	1.59	(0.3)
OND – ENG	3.54	(7.9)	4.46	(8.7)	8.74	(8.7)	9.83	(8.1)
PND – ENG	5.77	(10.3)	7.77	(12.43)	20.30	(18.7)	19.83	(12.1)
OND – GER	1.38	(3.0)	1.85	(4.2)	3.61	(4.6)	3.00	(4.7)
PND – GER	1.46	(4.4)	0.23	(0.6)	7.48	(12.7)	3.13	(4.2)
Rated concreteness	6.22	(0.4)	6.25	(0.3)	6.27	(0.5)	6.21	(0.3)

2.1.3. Procedure

Before the experiment proper started, instructions (in English) were displayed for each participant on the computer screen. The participants were informed that they would see English words (one by one) which they had to translate orally (as quickly and as accurately as possible into the microphone) into Polish or German (depending on the block). When translating into German, the participants were asked not to use articles. This decision was motivated by the fact that Polish does not use articles in unmarked contexts and Polish speakers of German as a foreign language often have problems with the correct article use for German nouns. Hence, avoidance of articles was required to eliminate confounds resulting from article selection. Next, the instructions were repeated to participants orally by the experimenter and the participants could ask questions. From the onset of instructions until all experimentation was over, the participants were addressed only in English. When the participants had no questions, they performed a practice session of 15 trials.

The following trial procedure was used during the training session and the experiment proper. First a fixation mark "x" appeared on the screen for 1000ms. Then, a blank screen was shown for 1000ms followed by the target (in capital letters) shown until a verbal response was given to the microphone or for max. 4000ms. When the response was registered by the voice key, the following message appeared on the screen "response detected" (in green ink). When no response was registered within the 4000ms time limit, "response not detected" appeared on the screen (in red ink). The feedback messages remained on the screen for 1500ms. Care was taken to limit the experimenter presence during the experiment in order not to intimidate the participants. Hence, the participants were tested individually in an sound-isolated booth (without any other person present during testing). Therefore, the feedback messages were meant to signal to the participants that their response was recorded by the equipment (but not whether it was correct or not). The participants had been informed about the aim of the messages and familiarised with them during the training session. The participants delivered their translations of English targets into German and Polish in 2 separate language blocks whose presentation was counterbalanced across participants. All stimuli in each of the blocks were presented in a randomised list. Each list contained half of all cognates, non-cognates and fillers so that each participant translated all targets; but each target was translated either into German or into Polish for a given participant. Training sessions preceded each language block in order to give time to the participants to feel comfortable with the switch of the translation

direction. All verbal responses given by the participants during the experiment were recorded and stored on the computer disk for further analysis of accuracy. The presentation of the stimuli and the recording of responses was controlled by E-Prime Professional (Schneider et al. 2002). After the experiment proper, the participants were asked to perform the English and the German versions of LexTALE and then answer the language background questions. The order of presentation of LexTALE language versions was counterbalanced across participants. The participants received course credits for their participation.

2.1.4. Data trimming and analysis

After the removal of responses to fillers, all verbal responses given by the participants were next transcribed by a proficient translator of Polish, German and English and next scored with respect to accuracy (1=accurate, 2=inaccurate, 0=omission). Verbal disfluencies, voice-key failures (including responses below 300ms), verbal gap fillers preceding the response or responses with articles were excluded from the analysis (3.25% of the responses). Inaccurate responses, i.e. responses with words which were not translation equivalents of the targets (6.99% of data) and omissions (20.44% of data) (henceforth errors-omissions) were next removed from the analysis of RT data. Next, from the analyses of RTs we excluded all responses exceeding 2.5xSD around the participant's mean (further 6.08% of the correct responses). Since most translations of the cognate VACUUM were STAUBSAUGER and ODKURZACZ making the target a non-cognate, all responses to VACUUM (and the matched non-cognate DONKEY) were excluded from the analysis. Altogether, 30.38% of all obtained responses were excluded from the RT analysis.

All data were analysed using LME models from the lme4 library (Bates et al. 2015) of the R system for statistical computing (R Core Team 2015). These models allow for analysing unbalanced designs where the number of observations per condition is not equal (as is the case in the present study); this type of analysis combines the traditional ANOVA F1 and F2 analyses by treating participants and items as random factors, which does not necessitate the aggregation of RTs over items or participants. Significance values for fixed effects were calculated using the lmerTest package of R (Kuznetsova et al. 2015). Sum contrasts were applied to the factors cognate status (cognates vs. non-cognates), item type (Polish–English vs. German–English) and translation direction (English–Polish vs. English–German), so that effect sizes reflected the distance of a condition from the grand mean. At the outset, all LME models

included a complex random factor structure for participants – the intercept and the slope for the cognate status, item type and the translation direction as well as the correlations between the intercepts and the slopes. Orthogonal to this, the model included the intercept and slope for the factor translation direction (English–Polish vs. English–German) for items. Moreover, all models included the typical word-level covariates – word length and word frequency counts (Zipf scale) as well as LexTALE scores for English and German (also as covariates) since our participants differed on those scores. Before the models were fitted, all covariates were mean-centred and the glmerControl(calc. derivs = FALSE) specification was added to the models. Application of a random factor principal component analysis (with the rePCA function), as suggested by Matuschek et al. (2017), indicated that this random factor structure was over-parameterized for all statistical models. To simplify the random factor structure, we first removed all random factor correlations, and when this did not eliminate over-parameterization, we removed the random factor component with the smallest amount of variance (Matuschek et al. 2017).

Due to the typical skewing of RTs toward longer durations, log transformations were applied to these data on the basis of the results of the BoxCox analysis. The resulting frequency distributions were close to normal. Effect sizes (b) and their standard error (SE) for analyses of RTs and accuracy are thus expressed in transformed units. For the sake of the clarity of presentation, reported means are based on the observed, rather than the transformed data. The translation accuracy data were analysed with generalized linear mixed models that converted the raw data to logits prior to analysis. Moreover, even though errors-omissions were not analysed as error rates (% errors), they are presented as such in tables in order to follow the psycholinguistic convention.

2.1.5. Results

The analysis of RTs revealed no main effect of cognate status ($b = -0.016$; $SE = 0.012$; $t = -1.36$; $p = .178$), nor an interaction of cognate status with item type ($b = -0.016$; $SE = 0.012$; $t = -1.344$ $p = .183$). This means that both types of cognates were translated as fast as non-cognates (no facilitation effect was recorded). However, we found a robust translation direction effect ($b = 0.231$; $SE = 0.0129$; $t = 18.00$, $p<.001$) – translation into Polish ($M = 1165$ ms; $SD = 437$) was 626 ms faster than translation into German ($M = 1791$ ms; $SD = 526$).

In contrast, the analysis of errors-omissions revealed a significant main effect of cognate status (b = 0.456; SE = 0.159; z = 2.865; p = .004) but the interaction of the cognate status with item type was insignificant (b = 0.04131; SE = 0.15926; z = 0.259; p = .795). This means that generally cognates were translated with greater accuracy than non-cognates but there was no notable difference in the size of the observed cognate advantage between Polish–English cognates and German–English cognates. Similarly, the size of cognate advantage in the accuracy data did not differ across translation directions (b = −0.10592; SE = 0.103; z = −1.027; p = .305). Yet we found a reliable main translation direction effect – translation into Polish was significantly more accurate than translation into German (b = −2.424; SE = 0.135; z = −17.99; p < 001). Mean RTs and error rates across conditions have been presented in Table 3. It needs to be noted that many of our participants voiced their comments after the experiment and said that they felt they had had to "mentally translate first into Polish and then into German" or that they "felt unable to translate from English into German as this was something we almost never do". Admittedly, such comments are only speculative and highly prone to error. However, they still suggest that the observed robust translation direction effect can point to a role of learning experience in language processing.

Table 3. Means RTs in ms (SD in parentheses) and mean error-omission rates (ER) (in % errors/omissions) across stimulus types and translation directions (POL=Polish, ENG=English, GER=German).

Stimulus type	Response language	Mean RTs (SD)	ER
POL–ENG cognates	POL	1300 (485)	15.5
non-cognates	POL	1334 (490)	29.0
POL–ENG cognates	GER	1740 (480)	54.0
non-cognates	GER	1861 (473)	61.9
GER–ENG cognates	POL	1079 (379)	7.1
non-cognates	POL	1099 (396)	13.3
GER–ENG cognates	GER	1801 (532)	25.4
non-cognates	GER	1775 (553)	36.7

Since data from Experiment 2 were collected before data from Experiment 1 were analysed, the discussion of the obtained results will be postponed until Experiment 2 has been reported.

2.2. Experiment 2

Since, in line with our predictions for the trilingual group in Experiment 1, we expected to find robust facilitation for Polish–English cognates but a significantly smaller or negligible facilitation for German–English cognates, the participants of Experiment 2 were originally expected to act as a control group. They were expected to exhibit a robust facilitation effect for German–English cognates since they concurrently used German and English during language learning. Undoubtedly, an ideal control group for the participants of Experiment 1 would comprise Polish learners of English who learned English via German, not Polish. However, it was impossible to find such a group as, within the Polish education system, Polish is typically used during foreign language instruction. Therefore, a group German learners of English in Austria who used German when learning English were asked to perform the translation task.

2.2.1. Participants

Forty-one German–English bilinguals participated in Experiment 2. They were all native speakers of the Austrian variety of German, who at the time of the experiment were students of English and American Studies, University of Vienna, Austria. The language background information (obtained via questions adapted from LHQ 2.0) revealed that none of the participants new Polish. Their LexTALE scores indicated that the participants were highly proficient speakers of English as L2. Details of participants' language profile are summarized in Table 4.

2.2.2. Materials

The materials used were identical to the ones used in Experiment 1.

Table 4. Basic characteristics of the participants in Experiments 2.

	English	
Self-reported proficiency – Listening	6.2	(0.6)
Self-reported proficiency – Reading	6.3	(0.7)
Self-reported proficiency – Speaking	5.8	(0.7)
Self-reported proficiency – Writing	5.9	(0.7)
LexTALE score	86.3	(10.2)
Age of onset of foreign language learning	10.2	(2.6)
Age	23.2 (2.7)	

2.2.3. Procedure

The procedure was very similar to the one employed in Experiment 1. The only exception was that here each participant translated both blocks of stimuli into German only.

2.2.4. Data trimming and analysis

After the removal of fillers, participants' responses were transcribed and categorised following the same procedures as the ones used in Experiment 1. Similarly to Experiment 1, verbal disfluencies, voice-key failures, verbal gap fillers preceding the response or responses with articles were excluded from the analysis (1.29% of the responses). Inaccurate responses (5.83% of data) and omissions (3.80% of data) were also removed from the analysis of RT data. Next, from the analyses of RTs we excluded all responses exceeding 2.5 × SD around the participant's mean (further 3.33% of the correct responses). Altogether, 13.90% of all obtained responses were excluded from the analysis of RTs.

The statistical analyses were similar to the ones in Experiment 1 but, since the participants of the current experiment translated the presented targets solely into German, the present analyses did not include the factor – translation direction. Since the analysis of latencies (but not accuracy) revealed a significant cognate status*item type interaction ($b = 0.037$; $SE = 0.014$; $t = 2.70$, $p = .009$), we performed further analyses in order to directly compare RTs to

Polish–English cognates vs. controls and to German–English cognates vs. controls. To that end, we recoded two factors – cognateness (cognates vs. non-cognates and item type (Polish–English cognates and controls vs. German–English cognates and controls) into a single factor stimulus category (Polish–English cognates vs. controls vs. German–English cognates vs. controls). We then used sliding contrasts (contr:sdif) to compare the neighbouring levels of the new factor. The maximal random structure for the new analyses included random slopes and intercepts for the factor stimulus category for subjects. Orthogonal to this, we included random intercepts for items. Similarly to Experiment 1, the models included centred covariates of word length and frequency as well as LexTALE scores for English and German. Model selection procedures were similar to those in Experiment 1 and were conducted in line with Matuschek et al. (2017). Since the models with the recoded factor were fitted to the same data, we applied the Bonferroni correction with significance levels at $p < .025$.

For the sake of the clarity of presentation, reported means are based on the observed, rather than the transformed data. As in Experiment 1, even though errors-omissions were not analysed as error rates (% errors-omissions), they are presented as such in tables to follow the psycholinguistic convention.

2.2.5. Results

The analysis of the RT data revealed a significant interaction of cognate status and item type ($b = 0.037$; $SE = 0.014$; $t = 2.701$; $p = .009$) but no other factors nor interactions reached significance (all $ps > .05$). As mentioned in section Data trimming and analysis, we subsequently performed further analyses to directly compare both types of double cognates to their matched controls. Those analyses showed that German–English cognates were translated significantly faster (136ms) than matched non-cognates ($b = 0.127$; $SE = 0.032$; $t = 3.895$, $p < .001$). Such a facilitation effect was not observed for Polish–English cognates ($b = -0.022$; $SE = 0.045$; $t = -0.493$; $p = .623$). The analysis of error-omission data obtained in Experiment 2 revealed no significant cognate effects nor interactions (all $ps > .05$). Since the analysis of RT data in Experiment 1 yielded null results with respect to cognate effects and the analysis of accuracy data revealed no cognate effects in Experiment 2, the two sets of findings are not directly comparable. Nevertheless, the juxtaposition of these two experiments seems to point to an important role of proficiency in processing, which

will be further elaborated on in the Discussion. The data obtained in Experiment 2 are summarized in Table 5.

Table 5. Means RTs in ms (SD in parentheses) and mean error-omission rates (in % errors omissions) across stimulus types obtained in Experiment 2 (POL = Polish, ENG = English, GER = German).

Stimulus type	Response language	Mean RTs (SD)	ER
POL–ENG cognates	GER	1237 (353)	22.0
non-cognates	GER	1178 (394)	19.2
GER–ENG cognates	GER	952 (273)	1.8
non-cognates	GER	1088 (337)	3.9

3. General discussion

The primary goal of the present study was to test the influence of learning experience on processing patterns of two types of double cognates relative to matched controls. We predicted robust cognate facilitation for cognates shared across two languages that had been concurrently used during language learning. In contrast, the facilitation for cognates shared across two languages that had not been concurrently used during learning was expected to be significantly weaker or negligible. The rationale for this pattern was based on the assumptions derived from RHM (Kroll et al. 2010; Kroll and Stewart 1994), which proposes that cognate facilitation emerges thanks to lexical links that are associative in nature (Dijkstra et al. 2010). Thus, the links should be strengthened with the increasing number of cross-language associations for words that are concurrently used during language learning but not for others.

To test these predictions, we compared the processing of Polish–English cognates vs. controls and German–English cognates vs. controls in two translation experiments. In Experiment 1 a group of Polish speakers of German and English (trilinguals) translated all stimuli into Polish and into German and in Experiment 2 a group of German speakers of English (bilinguals) translated all stimuli into German. Importantly, the trilinguals used Polish while learning German and English whereas the bilinguals used German while learning English. Consequently, we expected the trilinguals to display a robust cognate advantage for Polish–English cognates relative to matched controls but the effect for German–English cognates and controls was expected to be significantly smaller or even negligible. Furthermore, in line with the assumptions of RHM

for translation and cognate facilitation, the facilitation for Polish–English cognates should be larger during translation into Polish than during translation into German, whereas the facilitation for German–English cognates should display the opposite pattern. Simultaneously, we predicted that the bilingual group in Experiment 2 would exhibit significant facilitation effect only for German–English cognates relative to controls, as they knew no Polish and used German while learning English.

The secondary goal of this study was to investigate whether non-identical cognates would produce significant facilitation or inhibition in a translation task, given that there were no identical cognates in the stimulus list and that a translation task requires fundamentally different processing than an LDT (Comesaña et al. 2015). In the obtained data we found no evidence for a direct influence of learning experience on cognate processing. In contrast, a novel contribution of our study is a reliable facilitation effect (not inhibition) found for non-identical double cognates over non-cognates in the translation task performed by the trilingual group (in the accuracy data) and by the bilingual group (in the translation latencies).

The results of the present study revealed that trilinguals translated cognates with greater accuracy (but not speed) than non-cognates. However, these effects were not significantly different across the two cognate types or across the two translations directions, which did not match our predictions. Thereby we failed to find any support for the influence of learning experience on the cognate facilitation effect. These findings do not corroborate the currently postulated prediction within RHM (see Fig. 1) that lexical representations of words from the languages which were not concurrently used during learning are only weakly connected in the mental lexicons of language users. In contrast, we found a robust translation direction effect in trilinguals' RT data and in their error rates. They translated the English words significantly faster and with greater accuracy into the native language than into the other foreign language.

The present results obtained from the trilingual participants (Experiment 1) are at odds with the earlier findings reported by Lijewska and Chmiel (2015). Firstly, in contrast to the current experimentation, Lijewska and Chmiel found significant facilitation effect for German–English cognates relative to non-cognates when the stimuli were translated by Polish-German–English trilinguals into Polish (L1) but not when they were translated into German (L2).

Secondly, Lijewska and Chmiel reported the aforementioned pattern of cognate facilitation only for translation latencies as the analysis of accuracy

data was not included in their report. In contrast, we found a significant cognate facilitation (irrespective of translation direction and cognate type) in the accuracy data only (Experiment 1). Finally, Lijewska and Chmiel recorded smaller translation direction effect than the one we obtained in Experiment 1. Those discrepancies between the current results and those reported by Lijewska and Chmiel most likely stem from differences in the characteristics of the stimuli used in both studies as well as from different foreign language proficiency levels among participants of their study that in our trilingual group.

Lijewska and Chmiel combined identical and non-identical cognates in their stimulus list, which enabled them to employ more stimulus items within the same language combination. Consequently, more items tested increased power in their data and made the effects easier to detect in the collected RTs. Moreover, the inclusion of identical cognates in the stimulus list might have made the cognate facilitation itself stronger and more detectable. Previous research has shown that non-identical cognates tend to produce more subtle effects than identical cognates (Dijkstra et al. 2010; Van Assche et al. 2011); to the point that sometimes the facilitative effects of non-identical cognates may be undetectable (Duyck et al. 2007).

Interestingly enough, a closer analysis of the error rates Lijewska and Chmiel reported (but not analysed statistically) indicated that their accuracy data might actually be consistent with the significant cognate advantage currently observed for omissions and errors in Experiment 1. The numerical differences they reported seem to show that their German–English cognates might have been translated with greater accuracy than English-only words irrespective of the translation direction and (possibly) sentence context type. Unfortunately, this is just a speculation based solely on numerical values, as no statistical analyses of error data were given in Lijewska and Chmiel's report.

Nevertheless, we might have adopted too strict stimulus selection procedures. This, in combination with the typological differences between the tested languages (Schepens et al. 2012, 2013), lead to the generally lower number of items tested in our study than was the case for Lijewska and Chmiel's experiment. Furthermore, the procedures also caused an imbalance in the number of Polish–English cognates and controls relative to the German–English ones. The decision to select only concrete nouns in the present experimentation was taken on the basis of previous research indicating that concrete nouns tend to yield most reliable cognate effects (Bultena et al., 2013; Tokowicz and Kroll, 2007; Van Assche et al., 2013). Since other studies have demonstrated that cognate advantage tends to be reduced for non-identical cognates, we wanted to make sure that these subtle effects will not be diminished any further by the

variability within the selected stimuli. In retrospect, we probably should have included items from other word classes in order to increase power in our study. This limitation should be taken into account and addressed in future research. At present it cannot be excluded that the lack of evidence for the modulation of cognate advantage by learning experience is a consequence of power problems rather than a true reflection of the lack of such an influence in general.

Furthermore, the discrepancies between the results reported by Lijewska and Chmiel and those obtained in the current study likely stem from different proficiency levels among the tested participant groups. Even though our trilingual participants were recruited among 3rd and 4th year students of Applied Linguistics, whose classes, at the time of the experiment, were conducted mainly in German and in English, the obtained LexTALE scores indicated that the participants' foreign language proficiency was intermediate. This resulted in the exclusion of 30% of data due to omissions and errors, hence lowered the statistical power in the study. This, in turn, decreased the chances of observing reliable effects in RTs. In turn, Lijewska and Chmiel's participants were advanced speakers of both German and English. Past research with trilinguals indicates that some level of foreign language proficiency is required for subtle effects to be detectable (Szubko-Sitarek 2014; Van Hell and Dijkstra 2002). Therefore, the lack of significant cognate effects in translation latencies found in the present study might be a consequence of the generally lower proficiency of the tested participant sample and a lowered power due to the relatively high error rate (especially in English–German translation).

Further evidence for this interpretation comes from the juxtaposition of the data from Experiment 1 and from Experiment 2. We identified significant cognate facilitation in the accuracy data but not in RTs in Experiment 1, whereas we found significant facilitation for German–English cognates in RTs but not in the accuracy data collected from the bilingual group in Experiment 2 (as they most likely performed at ceiling level). Significant cognate effects were recorded in translation latencies obtained from advanced bilinguals in Experiment 2 and from advanced trilinguals in Lijewska and Chmiel's study, but not from the trilingual participants of Experiment 1 (whose German and English were intermediate). This seems to provide further evidence for the important role of proficiency in cognate processing (see Otwinowska (2015) and van Hell and Tanner (2012) for reviews).

Additionally, differences in the proficiency levels between trilinguals participating in the present study and the sample tested by Lijewska and Chmiel might have induced the robust discrepancy in the size of the observed translation direction effects. Their translation direction effect was more than two

times smaller than the one obtained in Experiment 1. Past research with bilinguals has shown that lower L2 proficiency leads to larger translation asymmetry (Francis et al. 2014; de Groot 1992; de Groot et al. 1994; Groot and Poot 1997; Van Hell and de Groot 1998). However, it needs to be noted that bilingual studies have demonstrated the influence of proficiency on translation asymmetry defined as the difference in the speed with which L1 words are translated into L2 and vice versa. The present study, especially when juxtaposed with data reported by Lijewska and Chmiel, contributes to this research and demonstrates that translation direction effect seems to be modulated by foreign language proficiency in trilinguals, too. This is the case when translation from a foreign language to the native one is compared to translation between two foreign languages.

Importantly, the emphasis on proficiency does not exclude the possibility that the large translation direction effect might have been additionally augmented by our participants' learning experience. It is possible that English–German translation was especially difficult for our participants as this was an atypical task for them. Such a perception of the task transpired from many comments the participants voiced once they completed the English–German translation task. When combined with the relatively low proficiency, the difficulty of the translation might have given rise to such a large difference between translation from English into German and the one from English into Polish. This issue definitely requires further research in order to be addressed more directly.

As far as the secondary aim of the present research is concerned, as predicted, we found significant cognate facilitation effects yielded by non-identical cognates in translation tasks performed by trilinguals and bilinguals, even when identical cognates were excluded from the stimulus list. Previous research with bilinguals found that non-identical cognates were recognised slower, rather than faster, than matched controls (Comesaña et al., 2015), when they were tested without any identical cognates in the stimulus list in L2 LDT. Comesaña et al. attributed the inhibitory effects generated by non-identical cognates to lateral inhibition of non-identical cognates competing for selection, which is not the case for single-language controls.

We predicted that the discrepancies between our RT data and Comesaña et al.'s findings most likely stem from differences in task demands between an LDT and a translation task. Comesaña et al. ascribed the cognate inhibition they obtained to the increased suppression of the non-target language in LDT when identical cognates were excluded from the task. As a consequence, strong lateral inhibition of non-identical cognates but not of controls lead to

slower processing of cognates over non-cognates. However, since such increased inhibition would be suboptimal during a translation task, which combines recognition in the source language with production in the target language, we predicted and subsequently found reliable facilitation rather than inhibition in the current task but only in accuracy data obtained from trilinguals and only in RTs produced by bilinguals. As has been mentioned earlier, this inconsistency in the observed cognate effects most likely stems from differences in proficiency levels between the tested groups but more research is needed in order to test this claim more directly.

4. Conclusions

Our novel finding was that non-identical cognates did produce facilitation in translation performed by trilinguals and by bilinguals, even when identical cognates were absent from the task. Furthermore, we found that significant cognate facilitation in the trilingual data was not modulated by cognate type or translation direction. Thereby, the obtained data did not provide direct evidence for the claim that how vocabulary is learned, i.e. which languages are concurrently used during foreign language learning, affects the subsequent cognate facilitation effects. Nevertheless, the obtained surprisingly large translation direction effect may suggest that learning experience affected translation in general, making translation in the unpractised direction an especially difficult task. In order to investigate this issue more directly and without the limiting factors of participants' language profile and the unbalanced design due to language typology, future research could possibly employ artificial languages/words to see how the language via which those items are learned subsequently affects their processing patterns.

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